

Lightweight Wearable Fingertip Haptic Device with Tangential Force Feedback based on Finger Nail Stimulation

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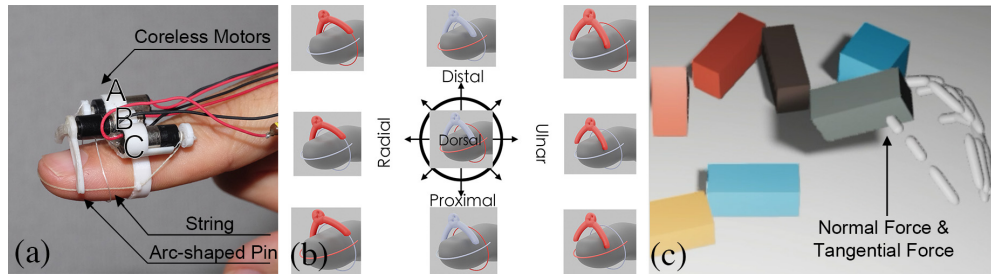


Figure 1: (a) The proposed haptic device features three motors controlling two strings (one for the fingertip, one for the finger pad) and an arc-shaped pin to stimulate the fingernail area and finger pad. Total weight: 5.24g. (b) Nine possible force patterns generated by the device: - Left column: Arc pin applies pressure on the finger's radial side with string-based drag forces. - Middle column: String-based tangential forces only, no pin pressure. - Right column: The arc pin applies pressure on the finger's ulnar side using string-based drag forces. (c) Virtual coupling hand and object manipulation.

ABSTRACT

We present a lightweight wearable haptic device that provides tangential force feedback through fingernail stimulation. The device features a ring-like structure mounted on the fingernail side, incorporating three miniature motors: two for string-based force feedback in the distal-proximal direction and one with an arc-shaped pin for radial-ulnar stimulation. By relocating the actuators to the nail side, our design preserves the natural tactile sensitivity of the finger pad while achieving 2-DOF tangential force feedback. The device weighs only 5.24g and can generate nine distinct force patterns through the combination of string tension and pin pressure. We conducted preliminary user studies to evaluate direction discrimination and the ability to perceive different weights and friction coefficients.

Index Terms: Virtual Reality, lightweight haptic display, tactile devices, dexterous manipulation, wearable haptics, 3DOF haptic display

1 INTRODUCTION

In virtual reality, tactile feedback - especially fingertip feedback - is crucial for dexterous object manipulation. While most current haptic devices focus on finger pad feedback, this approach has limitations: actuators placed on finger pads can interfere with real-world interactions, particularly in mixed reality settings, and also prevent users from using the device for a long time. This has led researchers to explore alternative methods for effective tactile feedback while minimizing device size and weight for better usability and comfort.

Tangential force is a key source of information about sliding, friction, and weight direction during object manipulation. When

the finger moves relative to an object, it primarily perceives the direction and magnitude of the frictional force. Objects of different masses will also exert a certain tangential component of gravity when in contact with the finger. With only normal force feedback, users may only perceive "touching an object" without truly sensing the lateral or downward force changes brought by the object. Tangential force can more clearly inform the user "from which direction the force comes" and "along which path it acts," simulating varying degrees of speed, acceleration, and other dynamic information. This is crucial for simulating operations such as dragging and stroking the surface of objects in dexterous manipulation.

Various haptic devices have been proposed to provide haptic feedback. Examples include glove-like systems offering distributed force feedback [5], handheld controllers that deliver localized pressure [1], and exoskeleton-style frameworks that lock or limit finger motion [3]. However, these devices can hinder hand dexterity. While many solutions provide normal force cues, adding tangential or multi-DOF feedback can be more complex. Some researchers attach multi-motor modules to the fingertip for directional force rendering [2], but this approach can obscure tactile sensations on the pad. To preserve the fingerpads' natural touch, researchers have developed alternatives that relocate actuators on the phalanx [4], or use soft thin-film actuators [7], remove the actuator when not in use [6]. However, these designs often limit normal or shear force output and also give the moveable part perceptible latency.

To address these challenges, the researchers propose a novel approach in this study that utilizes the fingernail as the primary location for tactile actuation. By shifting the actuator to the nail side, we aim to maintain the tactile sensitivity and dexterity of the finger pad, allowing for real-world interactions while providing precise and efficient directional feedback. Our design employs a ring-like structure to install three motors on the side of the fingernail shown in Fig. 1(a). Two motors are responsible for pulling strings to provide feedback in the distal and proximal directions. The other motor, B, is equipped with an arc-shaped pin designed to stimulate the radial and ulnar sides of the fingernail, providing tactile feedback along these two axes. This configuration ensures that users receive de-

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tailed feedback from multiple directions without compromising the natural interaction with physical objects. We selected planet-gear coreless motors for this device. These motors offer a balance of precision and power while remaining lightweight and small enough to integrate into the device without hindering user movement or comfort.

We conducted two user studies to evaluate the performance of the proposed device. The first study examined whether users could detect the directional feedback provided by the device, comparing scenarios with and without physical feedback. The second study assessed the perception of object weight and friction coefficient under three conditions, 3 DOF, 1 DOF, and vibration, to determine if these differences affected users' ability to discern tactile properties. Although our initial experiments were not perfect, some participants could correctly distinguish all 2DOF directions and differentiate between surfaces with varying friction and objects of different weights. The results of these experiments will be presented in future work once they are refined.

2 CONTRIBUTION

This study explores mechanical stimulation on the nail side to achieve directional tactile feedback and pressure sensation. Unlike previous work such as [9, 8], our approach achieves 2DOF tangential force feedback while maintaining the original structure and keeping the finger pad unobstructed without introducing additional types of actuators. By shifting the stimulation paradigm from the finger pad to the area near the fingernail, we explore a new possibility for haptic feedback that maintains device simplicity while enabling novel opportunities for tactile interaction design.

3 SYSTEM OVERVIEW

Our hardware setup includes a PIC32MK0512MCJ064-I/PT microcontroller for receiving haptic data via serial port signals and outputting up to 9 PWM channels to control multiple motors. We use the DRV8434e driver to regulate motor currents effectively. For actuation, we selected planetary gear coreless motors, each with a 6mm diameter and 12mm length. The arc-shaped pin motor operates at 1200 RPM, while the string-pulling motors function at 6400 RPM, providing sufficient pulling force for our application. We used a ring structure to mount three motors on the fingernail side. Motors A and C pull strings to provide distal and proximal feedback, while Motor B, with an arc-shaped pin, stimulates radial and ulnar sides for haptic feedback. Pressure in the dorsal direction is achieved by tightening Motors A and C together. See Fig. 1(b) for the control strategy, and Fig. 2 and Fig. 3 for the output force measurement.

In terms of software, we utilized Unity to create virtual environments and visual effects shown in Fig. 1(c). For the physics engine and high-fidelity haptic rendering, we used Springhead physics engine¹, which provides detailed information such as contact forces between rigid bodies, dynamic and static friction states and generates the stick-slip phenomena required for our experiment. We employed virtual coupling to construct virtual hands, which helps maintain system stability and ensures that the calculated feedback force is proportional to the penetration depth.

4 CONCLUSION AND FUTURE WORK

This paper presents a lightweight fingernail-mounted haptic device that provides directional tactile feedback through string-based actuation and pin stimulation. Our key contribution demonstrates that exploring a structure that stimulates the fingernail and finger pad can deliver 2DOF tangential force feedback while preserving

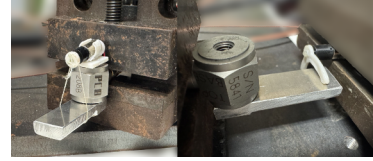


Figure 2: Experiment setup for the evaluation of drag force and pressure. On the left, the sensor is dragged by the string, and on the right, the pin presses the sensor.

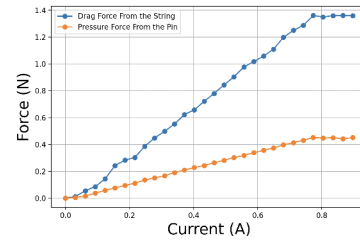


Figure 3: Drag force and pressure output regarding current. Both drag force and pressure show a linear relationship with current.

finger-pad interaction. Future work will focus on enhancing the mechanical design for more balanced force distribution and conducting extensive user studies to optimize the mapping between virtual interactions and the proposed feedback.

ACKNOWLEDGMENTS

This work was supported by JST SPRING, Japan Grant Number JPMJSP2180, and JSPS KAKENHI Grant Number 23H03432.

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¹<https://springhead.info/>